

ANDROID APPLICATION PROTOTYPE
MOBILE AUTHENTICATION & OPERATOR ACCESS (LOGINSCREEN)

MBPSAAS Mobile Application — Operator Login Screen (LoginScreen.kt)



MBPSAAS

Mobile-Based Poultry Security & Alert System

Username or Email:

user



Password:

.....



RETROFIT API READY

[Forgot Password?](#)

Login



Mobile Password Recovery — Step 1: Account Lookup (ForgotPasswordScreen.kt)

Step 1: Who are you?

Step 2: Security Question

Step 3: New Password

Enter your registered username or email to retrieve your security challenge question.

Username or Email:

user



[Back to Login](#)

Next Step →

ANDROID APPLICATION PROTOTYPE
PASSWORD RECOVERY CHALLENGE & PASSWORD RESET

Mobile Password Recovery — Step 2: Answer Security Question
(ExposedDropDownMenuBox)

Step 1: Who are you?

Step 2: Security Question

Step 3: New Password

Account Verified:

user

VERIFIED

Security Question:

What was your first pet's name?

▼

✓ What was your first pet's name?

What is your mother's maiden name?

What was the name of your elementary school?

What city were you born in?

Your Answer:

Bantay

[Back to Login](#)

Verify Answer →



Mobile Password Recovery — Step 3: Choose New Password & Confirmation Dialog

Step 1: Who are you?

Step 2: Security Question

Step 3: New Password

New Password:

.....

Confirm New Password:

.....

Password Updated (AlertDialog)

RESET SUCCESSFUL

Your password has been reset successfully. You can now log in with your new password.

[Back to Login Screen](#)

Reset Password & Continue

ANDROID APPLICATION PROTOTYPE
DASHBOARD SCREEN — NORMAL SAFE STATE (DASHBOARDSCREEN)

Overall Farm Security Banner — MotionStatusCard (Safe State)



ALL POULTRY ZONES SAFE

System Online — Monitoring active

PERIMETER CLEAR

Code Logic: `isMotionDetected = overallStatus == "MOTION_DETECTED" || latestEvent?.eventType == "MOTION_DETECTED" || latestEvent?.buzzerTriggered == true`. When false, uses SafeColor (#2E7D32) and primaryContainer.



Poultry Farm Zone Status Grid — ZoneStatusGrid (3-Zone Triplet)

Poultry Farm Zones (Real-Time Sensor Telemetry)



Coop Zone A
Clear

Brooder House



Coop Zone B
Clear

Layer Coop



Coop Zone C
Clear

Free-Range Pen

Mapping helper: `getZoneDisplayName("ROOMA")` → "Coop Zone A"

ALL ZONES SECURED

ANDROID APPLICATION PROTOTYPE
DASHBOARD SCREEN — INTRUSION ALERT STATE (DASHBOARDSCREEN)

Critical Intrusion Alert Banner — MotionStatusCard (Alarm Active)



INTRUSION ALERT — Coop Zone A

Last activity: 02:14:18 (Coop Zone A) • Siren Buzzer Active

BREACH DETECTED

Code Logic: Triggered when `overallStatus == "MOTION_DETECTED"` or `latestEvent?.buzzerTriggered == true`. Card color changes dynamically to `errorContainer` with `AlertColor` (#C62828).



Poultry Farm Zone Status Grid — Breach Indicator (Coop Zone A Triggered)

Poultry Farm Zones (Dynamic Visual Recomposition)



Coop Zone A
MOTION

Brooder House (PIR 1)



Coop Zone B
Clear

Layer Coop (PIR 2)



Coop Zone C
Clear

Free-Range (PIR 3)



Automated GSM SMS dispatched to Farm Admin

View Breach Details

ANDROID APPLICATION PROTOTYPE
PIR SENSORS ARMED / BYPASS CONTROLS (SensorControlCard)

Sensor Control Card — Interactive Multi-Zone Switches (SensorControlCard.kt)



PIR Sensors Control

Coop Zone A (Brooder House)

PIR Sensor 1 (Pin 12) • Active • Sounding Enabled

ARMED



Coop Zone B (Layer Coop)

PIR Sensor 2 (Pin 14) • Active • Sounding Enabled

ARMED



Coop Zone C (Free-Range Pen)

PIR Sensor 3 (Pin 13) • Bypassed for Maintenance

BYPASSED



Sensor State Synchronization & Backend API Dispatch Pipeline

1. User Action: Operator toggles Switch in Compose UI

INSTANT UI TOGGLE

2. Optimistic Update: `sensors = sensors.map { if (code == it.sensorCode) it.copy(isEnabled = new) }`

ZERO UI
LAG

3. HTTP Dispatch: `ApiClient.service.updateSensorStatus(code, isEnabled)`

RETROFIT POST

4. Rollback on Failure: If exception occurs, state safely reverts to previous server value

AUTOMATIC ROLLBACK

Endpoint: `/mbpsaas_api/update_sensor_status.php`

Arm All Sensors

ANDROID APPLICATION PROTOTYPE
 DASHBOARD SCREEN — RECENT INTRUSION ACTIVITY FEED

Recent Intrusion Activity Feed Component (DashboardScreen.kt)

Recent Intrusion Activity

Coop Zone A (Brooder House)
 2026-09-24 02:14:18 • Siren Buzzer Active

MOTION DETECTED

Coop Zone A (Brooder House)
 2026-09-24 02:14:45 • Hold 2000ms Cleared

MOTION STOPPED

Coop Zone C (Free-Range Pen)
 2026-09-24 01:42:10 • Siren Buzzer Active

MOTION DETECTED

Rendered via LazyColumn take(10) items
 [View All Triggered Alerts →](#)



MotionEventRow Component Anatomy & Data Model Breakdown

Model Field	Kotlin Data Type	Sample Value	UI Rendering Function
zone / zoneLabel	String / String?	"ROOMA" / "Coop Zone A"	Displays bold title via getZoneDisplayName()
eventType	String	"MOTION_DETECTED"	Selects Red Alert vs Green Safe pill badge
detectedAt	String	"2026-09-24 02:14:18"	Formats relative time • bodySmall caption
buzzerTriggered	Boolean	true	Displays acoustic alarm badge & sounds mobile audio

Performance Note: Rendered with Compose key(event.id) inside LazyColumn to guarantee smooth 60fps scrolling without unnecessary item recomposition during live 3s polling cycles.

ANDROID APPLICATION PROTOTYPE

TRIGGERED ALERTS — DATE & ZONE FILTERING (TRIGGEREDALERTSSCREEN)

Triggered Alerts Screen — Header Card & Date Filter Chips



Triggered Intrusion Alerts

14 alert logs recorded • Off-Hours Perimeter Breaches

Filter Alerts by Date:

All Dates

Today (Active)

Yesterday

Custom Date...

Filter Alerts by Zone:

All Zones (14)

Coop Zone A (8)

Coop Zone B (2)

Coop Zone C (4)



Android Native DatePickerDialog & Custom Date Filtering Flow

SELECT DATE

Thu, Sep 24, 2026

S	M	T	W	T	F	S
20	21	22	23	24	25	26

Cancel

OK

Implementation: Formats selected date into ISO prefix `selectedCustomDate = SimpleDateFormat("yyyy-MM-dd").format(cal.time)`. The Compose `remember(events, selectedDateFilter, selectedZoneFilter)` filter efficiently prunes the list without recomposing unaffected screens.

ANDROID APPLICATION PROTOTYPE

TRIGGERED ALERTS FEED & EMPTY VIEW STATES

Filtered Intrusion Alerts Feed — Chronological Alert Cards



Coop Zone A (Brooder House)

2026-09-24 02:14:18 • Continuous trigger 27s

ALERT TRIGGERED

Buzzer Active



Coop Zone C (Free-Range Pen)

2026-09-24 01:42:10 • Continuous trigger 19s

ALERT TRIGGERED

Buzzer Active

Filtered only events where eventType == "MOTION_DETECTED"

DELIVERED TO GSM



Empty Filter & Zero Intrusion Results View State



No Intrusion Alerts Found

There are no recorded intrusion alerts matching the selected date and zone filter. Your poultry farm is secure.

Reset Date Filter to All Dates

Loading State Note: When `isLoading == true`, a centered `CircularProgressIndicator()` is rendered in place of the list to provide responsive visual feedback during data fetching.

ANDROID APPLICATION PROTOTYPE
 ACTIVITY LOG STREAM & OPERATOR PROFILE MANAGEMENT

Activity Log Screen — Full Telemetry Stream (ActivityLogScreen.kt)

ROOMA: Coop Zone A • MOTION_DETECTED2026-09-24 02:14:18

ROOMA: Coop Zone A • MOTION_STOPPED2026-09-24 02:14:45

ROOMC: Coop Zone C • MOTION_DETECTED2026-09-24 01:42:10

Comprehensive audit record of all detection and clear transitions

LOGS SYNCHRONIZED



Operator Profile — ProfileScreen.kt (View & Edit In-Place)

User
 Farm Operator • user@poultryfarm.com

Edit Profile

Username:

user

Email:

user@poultryfarm.com

Current Password:

.....

New Password:

.....

Log Out of System

Cancel

Save Changes

SPECIFICATION & DESIGN SYSTEM
MOBILE ARCHITECTURE & COMPOSE DESIGN SYSTEM

Mobile Client Network Architecture & Telemetry Pipeline

- API CLIENT** **Retrofit Singleton (ApiClient.kt):** Configured with OkHttpClient and GsonConverterFactory; dual support for Wi-Fi Mode (http://10.192.10.14/mbpsaas_api/) and USB Mode (http://localhost:8080/mbpsaas_api/ via ADB reverse).
- POLLING** **3-Second Coroutine Loop (HomeScreen.kt):** Uses Kotlin coroutines with `delay(3000)` inside a lifecycle-aware `LaunchedEffect` to achieve pseudo-real-time poultry security telemetry.
- STATE** **State Hoisting & Recomposition:** Mutable states for events, zones, and sensors drive instant reactive updates across Dashboard, Alerts, and Activity Log tabs.
- ENDPOINTS** **Retrofit ApiService.kt:** `login.php`, `forgot_password.php`, `get_motion_events.php`, `get_sensors.php`, `update_sensor_status.php`, and `update_profile.php`.



Uniform Material 3 Compose Color System & Design Specifications

 Primary Green #1b5e20	 SafeColor #2e7d32	 AlertColor #c62828	 DisabledColor #757575	 InfoColor #0288d1	 OnSurface Ink #102a16
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Mobile UI Directives: All screens follow standard Android Material 3 mobile design paradigms. Interactive buttons use Primary (#1b5e20) or SafeColor (#2e7d32) with minimum 48dp touch targets. Critical intrusion alerts and warnings strictly utilize AlertColor (#c62828). Normal safe conditions utilize SafeColor (#2e7d32). Monospaced typography applies to all endpoint URLs, token strings, and zone codes.